

CHEMICAL EQUILIBRIUM

Forward Reaction: The reaction in which reactants react to form Products is called forward Reaction.

→ It takes place from left to right.

Example: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

Backward Reaction: The reaction in Which products react to produce reactants is called backward reaction.

→ It takes place from right to left.

Example: $2\text{NH}_3 \rightarrow (\text{g}) \text{N}_2 + 3\text{H}_2(\text{g})$

Chemical Equilibrium: In case of a reversible reaction the reactants react with one another to produce Products. And, On the other hand these products reacts with each other and form reactants back. So in these case, when the reaction among the reactants Proceeds, after some time the rate of reaction of the reactants to Products equals to the rate of reaction of the Products to Produce the reactants back. The state is called chemical equilibrium.

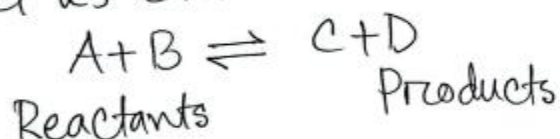
Characteristics of chemical equilibrium: There are some characteristics of Chemical equilibrium. They are –

1. **Reversibility of Reaction:** Chemical equilibrium is related only to reversible reactions. It is not applicable to irreversible reaction.
2. **Stability of equilibrium:** If the external Conditions (Pressure, temperature and concentration) are not changed, the system will stay for ever.
3. **Easy approachability from both sides:** The equilibrium can be attained from both directions.
4. **Closed system:** It is only attained in Closed system.
5. **Ineffectiveness of Catalysts:** The Catalysts have no effect on chemical equilibrium

Law of mass action: In 1867 Mathematician C.M. Guldberg and Chemist P. Waage of Norway invented a law regarding the rate of chemical reactions, which became Known as law of mass action. The law is as follows "The rate of a Chemical reaction is Proportional to the active masses i.e. Concentration and Partial Pressure) of the reactants.

⇒ Equilibrium Constant:

Let us consider following general reaction.



According to the law of mass action, the rate of forward reaction,

$$R_f \propto [A][B]$$

$$R_f = K_1 [A][B]$$

Where, $[A]$ & $[B]$ are the molar concentration of the reactants A & B respectively.

K_1 = Rate constant of the forward reaction

Similarly, the rate of backward reaction,

$$R_b \propto [C][D]$$

$$R_b = K_2 [C][D]$$

At the equilibrium, the rate of forward reaction and backward reaction remain same.

$$R_f = R_b$$

$$K_1 [A][B] = K_2 [C][D]$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C][D]}{[A][B]}$$

$$\therefore K_c = \frac{[C][D]}{[A][B]}$$

Where,

$[C]$ = Molar concentration of the product C.

$[D]$ = Molar concentration of product D.

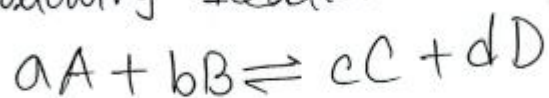
K_2 = Rate constant of the backward reaction.

Here, K_c is called molar equilibrium constant or simply equilibrium constant of the reaction.

$$\therefore K_c = \frac{[C][D]}{[A][B]}.$$

This is the mathematical expression for the law of chemical equilibrium.

✶ Write the K_c & K_p value for the following reaction.



Forward Reaction Rate,

$$R_f \propto [A]^a \times [B]^b$$

$$\therefore R_f = K_1 [A]^a [B]^b$$

Backward Reaction Rate,

$$R_b \propto [C]^c [D]^d =$$

$$R_b = K_2 [C]^c [D]^d$$

At equilibrium,

$$R_f = R_b$$

$$K_1 [A]^a [B]^b = K_2 [C]^c [D]^d$$

$$\text{or, } \frac{K_1}{K_2} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\text{or, } K_c = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b}$$

$$\text{And, } K_p = \frac{P_C^c \times P_D^d}{P_A^a \times P_B^b}$$

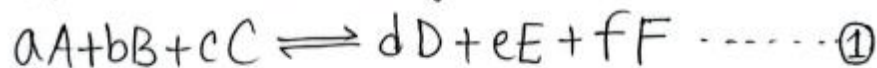
Where, K_c = Equilibrium constant at molar concentration

K_p = Equilibrium constant at Partial Pressure.

Q. Establish the relation between K_p and K_c for the given reaction.



Let us consider the given reversible reaction,



The rate of forward reaction,

$$R_f \propto [A]^a [B]^b [C]^c$$

$$R_f = K_1 [A]^a [B]^b [C]^c \dots\dots\dots ②$$

The rate of backward reaction,

$$R_b \propto [D]^d [E]^e [F]^f$$

$$R_b = K_2 [D]^d [E]^e [F]^f \dots\dots\dots ③$$

At equilibrium,

$$R_f = R_b$$

$$\text{or, } k_1 [A]^a [B]^b [C]^c = k_2 [D]^d [E]^e [F]^f$$

$$\text{or, } \frac{k_1}{k_2} = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c}$$

$$\text{or, } K_c = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c} \dots\dots (4)$$

$$\text{or, } K_c = \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \dots\dots (5)$$

$$\text{or, } K_p = \frac{P_D^d \times P_E^e \times P_F^f}{P_A^a \times P_B^b \times P_C^c} \dots\dots (6)$$

Now, we know from ideal gas's equation,

$$PV = nRT$$

$$\text{or, } P = \frac{n}{V} RT \quad [C = \frac{n}{V}]$$

$$\text{or, } P = CRT \dots\dots (7)$$

Putting the value of 'P' in eqn (6), we get

$$K_p = \frac{(CRT)_D^d \times (CRT)_E^e \times (CRT)_F^f}{(CRT)_A^a \times (CRT)_B^b \times (CRT)_C^c}$$

$$\text{or, } K_p = \frac{C_D^d \times (RT)^d \times C_E^e \times (RT)^e \times C_F^f \times (RT)^f}{C_A^a \times (RT)^a \times C_B^b \times (RT)^b \times C_C^c \times (RT)^c}$$

$$\text{or, } K_p = \frac{C_D^d \times C_E^e \times C_F^f \times (RT)^{d+e+f}}{C_A^a \times C_B^b \times C_C^c \times (RT)^{a+b+c}}$$

$$\text{or, } K_p = K_c \times (RT)^{(d+e+f)-(a+b+c)} \quad [\text{from eq}^n \text{ (5)}]$$

$$\text{or, } K_p = K_c \times RT^{(n_2 - n_1)}$$

$$\therefore K_p = K_c \times (RT)^{\Delta n}$$

Where, Δn = Total mole number of products (n_2)
 — total mole number of reactants (n_1)

Le-Chatelier Principle:

"If a change occurs in one of the controlling factors, such as temperature, pressure, concentration etc. under which a system is in equilibrium the system will tend to adjust itself in such a way so as to reduce the effect of that Change."

In other words,

"When a chemical reaction at equilibrium is subjected to a stress, the system will shift the equilibrium point in the direction that counteracts the stress applied"

There are three ways in which the stress can be caused on a chemical equilibrium:

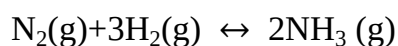
1. Changing the concentration of a reactant Or Product
2. Changing the Pressure (on volume) of the system.
3. Changing the temperature.

if a change in concentration, Pressure or temperature is caused to a chemical reaction in

equilibrium, the equilibrium will shift to the right or the left so as to minimize the change.

Effect of change in concentration:

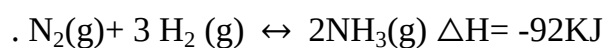
When concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.



When N_2 OR H_2 is added to the eqm Stage, the equilibrium will shift to the right side so as to reduce the concentration of the N_2 OR H_2 .

Effect of change of temperature:

When temperature of a reaction is increased the eqm shifts in a direction in which heat is absorbed.

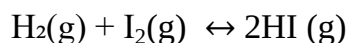


an exothermic reaction and When heat is increased in this reaction. the equilibrium will shift to the left. Therefore, NH_3 Production will be decreased.

Effect of change of pressure:

When Pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.

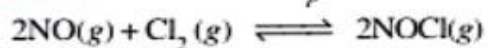
The chemical equilibrium in which no difference both total number of mole of reactants and total number of mole of Products and the equilibrium isn't a gaseous equilibrium, there is no effect of Pressure on equilibrium.



of No effect Pressure on equilibrium

$$= 1.9 \times 10^3$$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ or } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

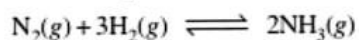
$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

$$\begin{aligned} K_c &= \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}} \\ &= 4.6 \times 10^4 \end{aligned}$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2 , 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction



at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[N_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$

$$[H_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$

$$[NH_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c .

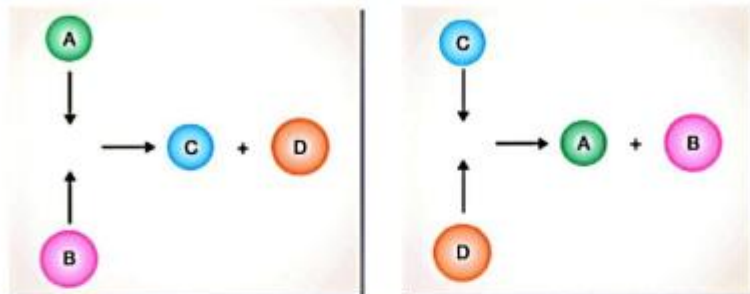
$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C, we can write

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \mathbf{0.080}$$

Chemical Equilibrium is Dynamic Equilibrium

We have shown above that as the reaction $A + B \rightleftharpoons C + D$ attains equilibrium, the concentrations of A and B, as also of C and D remain constant with time. Apparently it appears that the equilibrium is dead. But it is not so. The equilibrium is dynamic. Actually, the forward and the reverse reactions are taking place at equilibrium but the concentrations remain unchanged.



■ **Figure 17.2**
Molecules of A and B colliding to give C and D, and those of C and D colliding to give A and B.

624 **17** PHYSICAL CHEMISTRY

The dynamic nature of chemical equilibrium can be easily understood on the basis of the kinetic molecular model. The molecules of A and B in the equilibrium mixture collide with each other to form C and D. Likewise C and D collide to give back A and B. The collisions of molecules in a closed system is a ceaseless phenomenon. Therefore collisions of A and B giving C and D (Forward reaction) and collisions of C and D giving back A and B (reverse reaction) continue to occur even at equilibrium, while concentrations remain unchanged.

CHEMICAL KINETICS

RATE OF REACTION : Chemical kinetics may also be called reaction kinetics or simply "kinetics". The rate of a chemical reaction usually has units of sec^{-1} . The reaction rate is the rate at which the reactants of a chemical reaction form the products. Reaction rates are expressed as concentration per unit time.

Zero order Reaction: The reaction whose rate is independent of the concentration of a reactant, so that changing its concentration has no effect on the speed of the reaction is called zero-order reactions.

Example: $(i) 2 \text{NH}_3(\text{g}) \rightarrow 3 \text{H}_2(\text{g}) + \text{N}_2(\text{g})$

Pseudo unimolecular reaction Definition: It is a reaction which seems to be second degree but actually a first order reaction. One of the reactant is in excess which makes its importance negligible and to follow the first order rate is called, "Pseudo Unimolecular Reaction".

Example: $(i) \text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} (\text{excess}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{C}_6\text{H}_{12}\text{O}_6$ Rate = $K [\text{C}_{12}\text{H}_{22}\text{O}_{11}]$

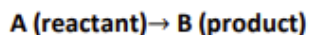
$(ii) \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} (\text{excess}) \rightarrow \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$ Rate = $K [\text{CH}_3\text{COOC}_2\text{H}_5]$

Order & Molecularity

Molecularity	Order
Molecularity of a reaction is the number of atoms, ions or molecules taking part in a reaction as in chemical equation.	Order of reaction is the total power of concentration of reactants, which is proportional to the experimental facts.
Molecularity is a theoretical concept which depends on chemical equation partly.	Order of reaction is an experimental fact which depends on experiment, observation and conclusion.
Molecularity can be fraction	Order of reaction never be fraction
- Example: $2 \text{NH}_3(\text{g}) \rightarrow 3 \text{H}_2(\text{g}) + \text{N}_2(\text{g})$ - Molecularity=2	- Example: $2 \text{NH}_3(\text{g}) \rightarrow 3 \text{H}_2(\text{g}) + \text{N}_2(\text{g})$ - Order- Zero order reaction

First order reaction: A reaction in which the first power of concentration term enters into the general equation is called first order reaction.

Mathematical Expression: Let us consider the first order reaction as follows :



Initially : $a \text{ mol L}^{-1}$ 0

After time, t : $(a-x) \text{ mol L}^{-1}$ x

If the initial concentration of reactant A is 'a' mol and after time 't' x mol of A has been converted to product B then $(a - x)$ mol of A remains. So according to Law of mass action, rate of first order reaction at that time will be :

$$\frac{dx}{dt} = k \times (a - x)^1$$

or, $\frac{dx}{(a-x)} = k \times dt$, here k is a constant.

On integrating both sides we get:

$$\int \frac{dx}{(a-x)} = \int (k \times dt),$$

or, $-\ln(a-x) = k \times t + c$ (integration constant)

When, $t = 0$, then $x = 0$, so $-\ln a = c$ (const). Putting this value in equation (1) we get.

$$-\ln(a-x) = k \times t - \ln a \text{ or } k \times t = \ln \frac{a}{a-x}$$

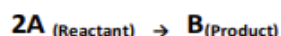
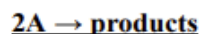
$$\text{Or, } k = \frac{1}{t} \times \ln \frac{a}{a-x}$$

or, $k = \frac{2.303}{t} \times \ln \frac{a}{a-x}$ this is the rate constant of a first order reaction.

(A) **Unit of k of first order reaction :** From the equation for rate constant k of a first

order reaction: $k = \frac{2.303}{t} \times \ln \frac{a}{a-x}$, we get unit of $k = s^{-1}$ or min^{-1}

- **2nd Order reaction:** - The reaction whose rate is proportional to the square of concentration of one reactant or to the concentrations of two reactants is called a second order reaction.
- **Rate equation for 2nd order reaction: let us consider the following 2nd order reaction,**



Initially, t = 0, a moleL⁻¹ → 0

After time, t = t, (a-x) moleL⁻¹ → x

If the initial concentration of A be a mol L⁻¹ and if in time “t” the concentration of A will be (a-x) mol/L. So according to law of mass action at any time t the rate of reaction will be:

$$\frac{dx}{dt} \propto (a-x)^2$$

$$\Rightarrow \frac{dx}{dt} = k (a-x)^2$$

$$\Rightarrow \frac{dx}{(a-x)^2} = k dt$$

$$\Rightarrow \int \frac{dx}{(a-x)^2} = \int k dt$$

$$\Rightarrow \frac{1}{a-x} = kt + C \dots \dots \dots (i)$$

When, t=0, x=0 then from eqⁿ (i) we get,

$$\frac{1}{a} = 0 + C$$

$$\Rightarrow C = \frac{1}{a} \dots \dots \dots (ii)$$

Putting the value of C in eqⁿ (i) we get,

$$\frac{1}{a-x} = kt + \frac{1}{a}$$

$$\Rightarrow kt + \frac{1}{a} = \frac{1}{a-x}$$

$$\Rightarrow kt = \frac{1}{a-x} - \frac{1}{a}$$

$$\Rightarrow kt = \frac{a-a+x}{a(a-x)}$$

$$\Rightarrow kt = \frac{x}{a(a-x)}$$

$$\Rightarrow k = \frac{1}{t} \cdot \frac{x}{a(a-x)}$$